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Décomposition de HODGE–DE RHAM sur une surface fermée de $\mathbb{R}^3$: discrétisation DDFV

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Résumé

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Mots-clés.

Abstract

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Keywords.

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PREMIÈRE PARTIE

Introduction

Chapitre I

Inégalité de POINCARÉ DDFV sur une surface fermée sans bord de $\mathbb{R}^3$

Sommaire du chapitre

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I.1 Generalization of the DDFV method to 2D surfaces in $\mathbb{R}^3$ [TMR16 ; TM19] [ABH07, Sec. 2.1]

Construction maillages.

1. Soit une surface $\mathcal{M}$
2. On place des points sur la surface $\mathcal{M}$
3. On en déduit un complexe sur la variété
4. Ce complexe définit le maillage primal, courbe et polyédrique
5. On calcule les barycentres des mailles primales plates
6. On projette ces barycentres sur la surface $\mathcal{M}$
7. Construction des mailles duales (barycentriques):

8. Projection du milieu des arêtes primales sur $\mathcal{M}$
9. Arcs passant par les milieux projetés
10. Les mailles duales sont approchées par $2m$ facettes

Let $\varphi: M \rightarrow \mathbb{R}^3$ be a smooth embedding of a 2-dimensional compact (boundaryless) manifold M in Euclidean space.

I.1.1 Primal mesh

The construction of a mesh for DDFV method is based on the triangular mesh used in cotangent scheme [Mey+03]. It considers a triangulation $\mathbf{M}$ of the manifold M , which is a simplicial complex homeomorphic to M . Corresponding homeomorphism $\rho: \mathbf{M} \rightarrow M$ induces a triangular structure on M consisting of vertices $\chi = \rho(x)$, edges ς and triangles K , where x are the vertices of the triangulation $\mathbf{M}$. The surface $\mathcal{M} \stackrel{\text{def.}}{=} \varphi(M)$ is also endowed with a triangular structure induced by the map $\varphi \circ \rho$ (I.1, bottom right).

An approximation of $\mathcal{M}$ is defined using a new embedding $\bar{\varphi}: M \rightarrow \mathbb{R}^3$ which is linear on each triangle $K \in M$. The surface $\mathcal{M}_h \stackrel{\text{def.}}{=} \bar{\varphi}(M)$ is a polyhedral approximation of $\mathcal{M}$ (I.1, bottom left) with vertices $x \stackrel{\text{def.}}{=} \varphi(\chi) = \bar{\varphi}(\chi)$, edges $\sigma = \bar{\varphi}(\varsigma)$ and triangular faces $\kappa = \bar{\varphi}(K)$. The *primal mesh* $\mathfrak{M}$ is the set of disjoint polygonal control volumes κ .

We create a representative point x_κ of each triangle $\kappa \in \mathfrak{M}$ (I.1, bottom left). The point x_κ does not need to lie in the triangle, only its projection to the hyperplane containing the triangle κ does. We call $\mathfrak{B}$ this family of points and let $\mathfrak{B}^*$ denote the set of vertices of the mesh $\mathfrak{M}$. For all adjacent control volumes κ and $\mathcal{L}$, we assume that $\partial\kappa \cap \partial\mathcal{L}$ is a segment that we call *an edge of the mesh* $\mathfrak{M}$ and that we denote by $\sigma = \kappa \mid \mathcal{L}$. Let $\mathfrak{E}$ be the set of such edges. For all adjacent control volumes κ and $\mathcal{L}$, we denote by $\sigma^* = (x_\kappa, x_\mathcal{L})$ the segment joining their representative points. Let $\mathfrak{E}^*$ be the set of such segments.

I.1.2 Diamond set

Next, we construct a diamond set as follows. We consider a couple $(\sigma, \sigma^*) \in \mathfrak{E} \times \mathfrak{E}^*$ such that $\sigma = \kappa \mid \mathcal{L}$ and $\sigma^* = (x_\kappa, x_\mathcal{L})$. We denote by x_{κ^*} and $x_{\mathcal{L}^*}$ the endpoints of the edge σ . We consider the plane P_{σ, σ^*} passing through the midpoints of the segments (x_κ, x_{κ^*}) , $(x_\kappa, x_{\mathcal{L}^*})$, $(x_\mathcal{L}, x_{\mathcal{L}^*})$ and $(x_\mathcal{L}, x_{\kappa^*})$ (I.2a sketched in black). We then define the quadrilateral diamond cell $\mathcal{D}_{\sigma, \sigma^*}$ whose vertices are the projections of the points x_κ , $x_\mathcal{L}$, x_{κ^*} and $x_{\mathcal{L}^*}$ onto the plane P_{σ, σ^*} in the direction of the vector $\sigma \times \sigma^*$ (I.2a). By abuse of notation, we still denote by σ and σ^* the diagonals of the diamond cell $\mathcal{D}_{\sigma, \sigma^*}$, as shown on I.2b. A diamond is also illustrated on I.1, bottom left. Notice that the diamond cells are the union of two disjoint triangles and can be non convex. The set of diamond cells is denoted by $\mathfrak{D}$. Our diamond set is a generalization of 2D planar diamond set used in DDFV method developed in [DO05] to 2D surfaces in $\mathbb{R}^3$.

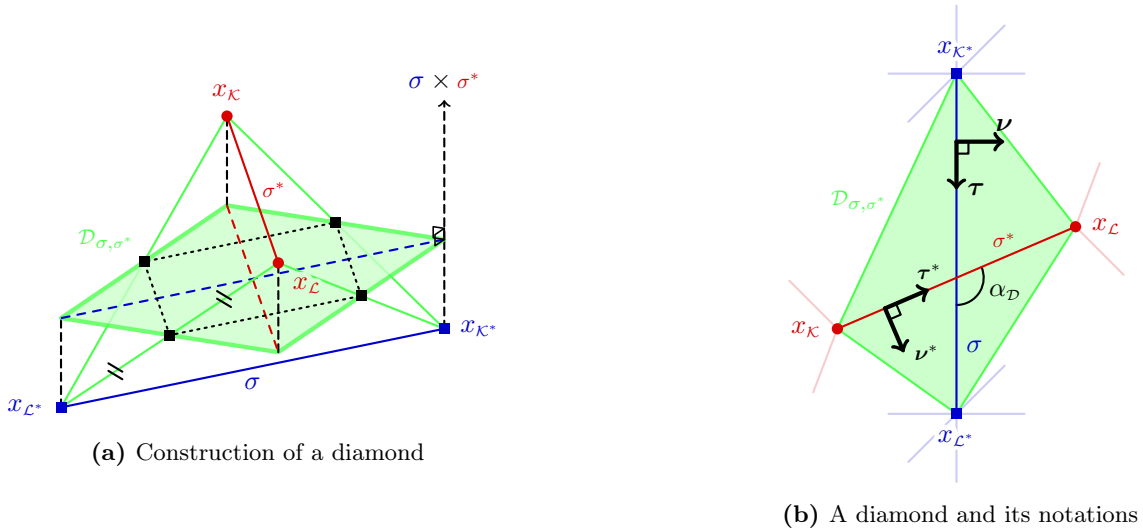


FIGURE I.2. Diamond

Remarque I.1.1. The plane P_{σ, σ^*} approximates the tangent plane to the surface $\mathcal{M}$ at ?.

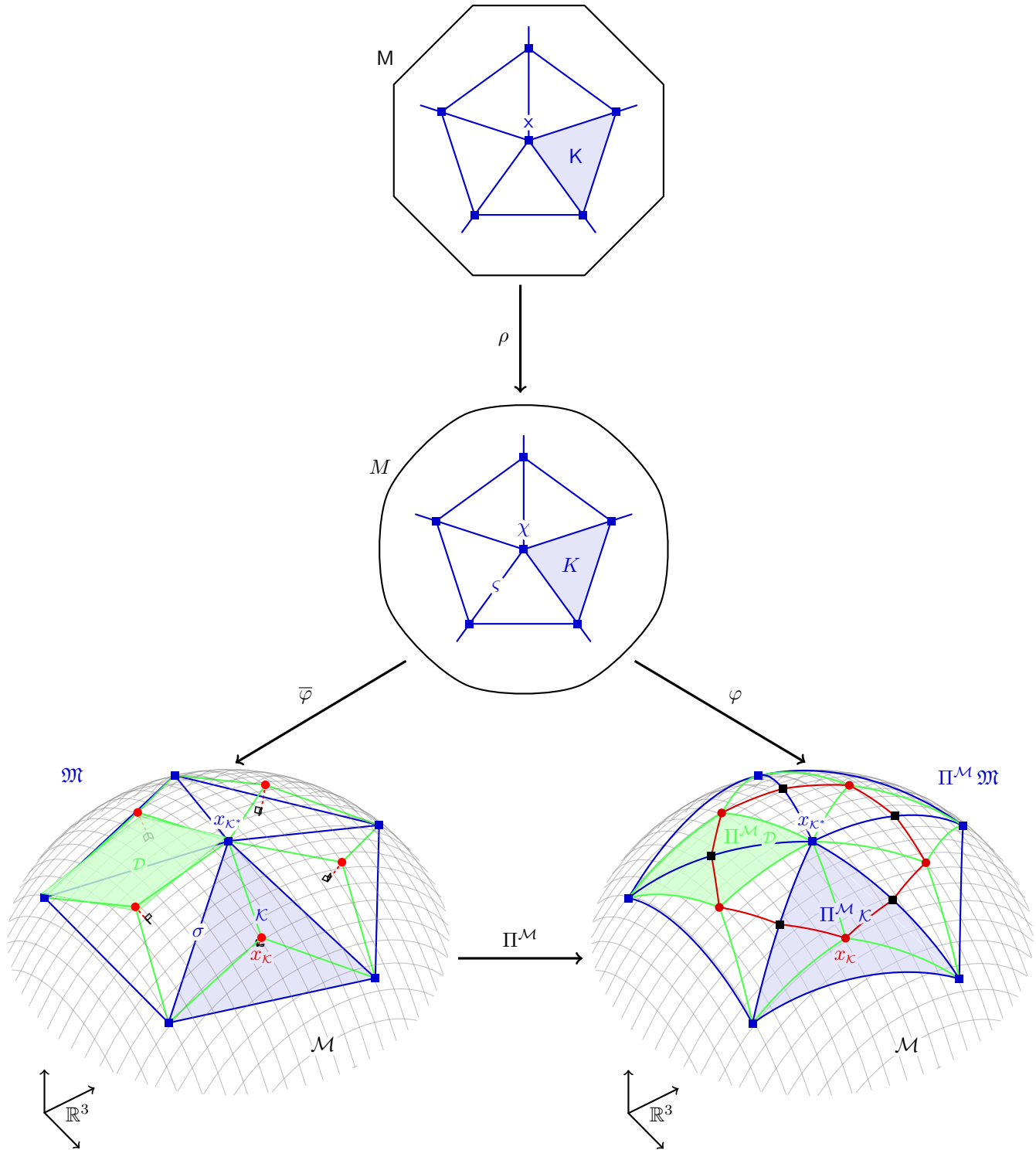


FIGURE I.1. Triangulated surface from the point of view of the differential geometry

Remarque I.1.2. As shown on I.3, the set of diamond cells $\mathfrak{D}$ does not constitute a mesh of the polyhedral surface $\mathcal{M}_h$.

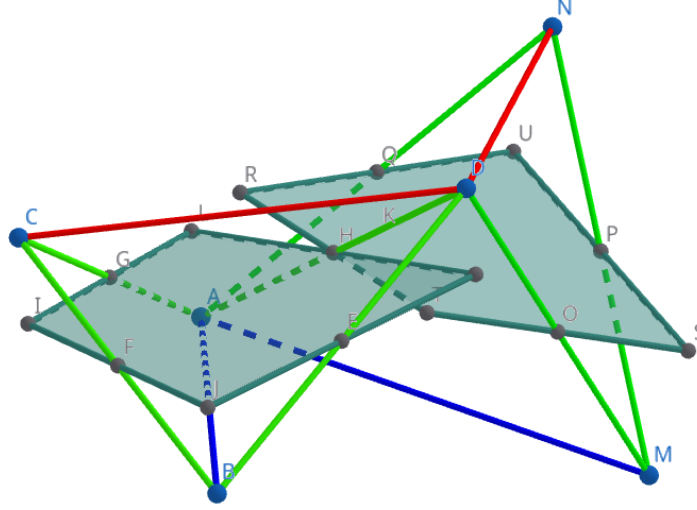


FIGURE I.3. Adjacent diamonds

Proposition I.1.3.

$$m(\mathcal{D}) = \frac{1}{2} \|\sigma \times \sigma^*\|.$$

$$m(\mathcal{D}) = \frac{1}{2} m(\sigma) m(\sigma^*) \sin(\alpha_{\mathcal{D}})$$

Bented diamond

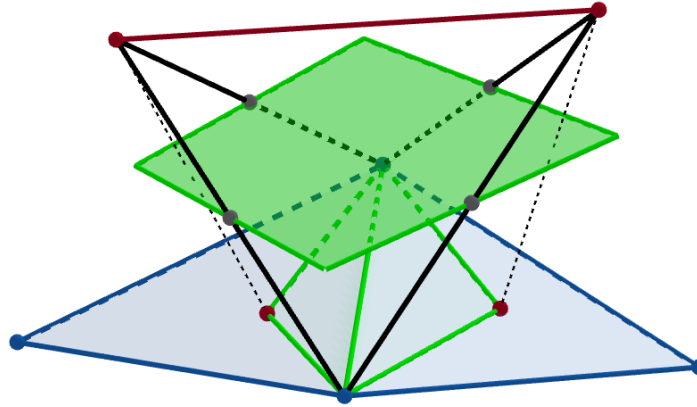


FIGURE I.4. A bented diamond

[Kre10] À chaque diamant $\mathcal{D} \in \mathfrak{D}$, on associe des demi-diamants de la manière suivante : $\tilde{\mathcal{D}}_p \stackrel{\text{d\'ef.}}{=} \mathcal{D} \cap p$ tel que $p \cap \mathcal{D} \neq \emptyset$ pour $p \in \{\mathcal{K}, \mathcal{L}\}$. On a $\mathcal{D} = \tilde{\mathcal{D}}_{\mathcal{K}} \cup \tilde{\mathcal{D}}_{\mathcal{L}}$. L'ensemble des demi-diamants est noté $\tilde{\mathfrak{D}}$. Pour chaque diamant $\mathcal{D} \in \mathfrak{D}$, on définit un sous-ensemble de demi-diamant de $\mathcal{D}$: $\tilde{\mathfrak{D}}_{\mathcal{D}} = \{\tilde{\mathcal{D}} \in \tilde{\mathfrak{D}} \mid \tilde{\mathcal{D}} \subset \mathcal{D}\}$.

Définition I.1.4 (Pseudo-indicatrice sur les diamants). Soit $\mathcal{D} \in \mathfrak{D}$. Soit $\mathbb{P}$ la projection de $\tilde{\mathfrak{D}}_{\mathcal{D}}$ sur $\mathcal{M}_h$. On

définit la fonction

$$\chi_{\mathcal{D}}: \begin{array}{l} \mathcal{D} \longrightarrow \{0, 1\} \\ x \longmapsto \mathbf{1}_{\mathbb{P}\tilde{\mathcal{D}}_{\mathcal{D}}}(\mathbb{P}x) \end{array} .$$

I.1.3 Dual mesh

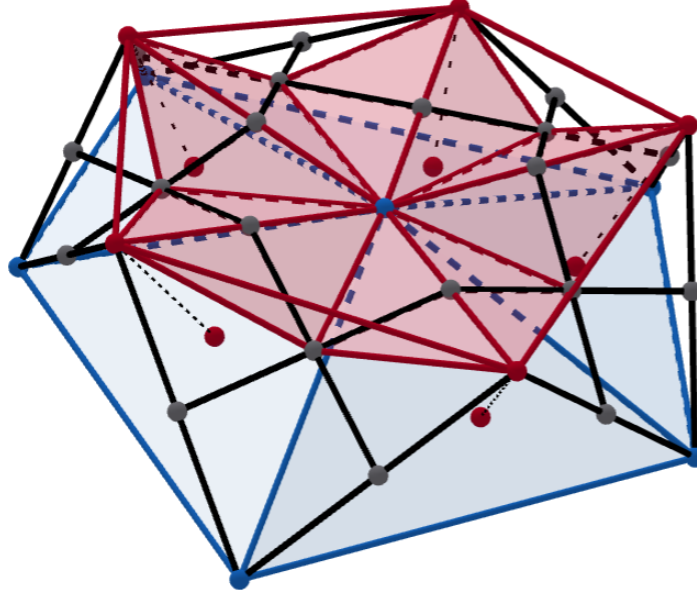


FIGURE I.5. Dual cell

The vertices $x_{\mathcal{K}^*}$ and $x_{\mathcal{K}}$ will be referred to as $\mathfrak{M}$ -vertices and $\mathfrak{M}^*$ -vertices respectively. The set $\mathfrak{M}^*$ is a family of «dual» control volumes defined as follows. To any point $x_{\mathcal{K}^*} \in \mathfrak{B}^*$, we associate the polygon $\mathcal{K}^* \in \mathfrak{M}^*$ whose vertices are $\{x_{\mathcal{K}} \in \mathfrak{B} \mid x_{\mathcal{K}^*} \in \bar{\mathcal{K}}, \mathcal{K} \in \mathfrak{M}\}$ sorted with respect to the clockwise order of the corresponding primal control volumes.

The corresponding notations $\sigma^* = \mathcal{K}^* \mid \mathcal{L}^*$ and $\mathfrak{E}^*$ refer to the dual mesh $\mathfrak{M}^*$.

The finite volume mesh is constructed by the barycentric subdivision of $\mathcal{M}$. The finite volume (or covolume) $V_i \mathcal{K}^* \in \mathfrak{M}^*$ around a vertex $\mathbf{X}_i x_{\mathcal{K}^*}$ is a region bounded by the piecewise linear curve joining the barycenters of the neighbouring triangles and the midpoints of the edges leading to the neighbouring vertices $X_{i,p}$, $p \in \llbracket 1; m \rrbracket$ where m is the number of neighbours.

I.1.4 DDFV mesh

We call $\mathcal{T}$ a double $(\mathfrak{M}, \mathfrak{M}^*)$ of meshes on the polyhedral surface $\mathcal{M}_h$, where $\mathfrak{M}$ is the *primal mesh* and $\mathfrak{M}^*$ is the *dual mesh*. Even though more general situations can be handled, we concentrate in this paper to meshes satisfying the following assumption.

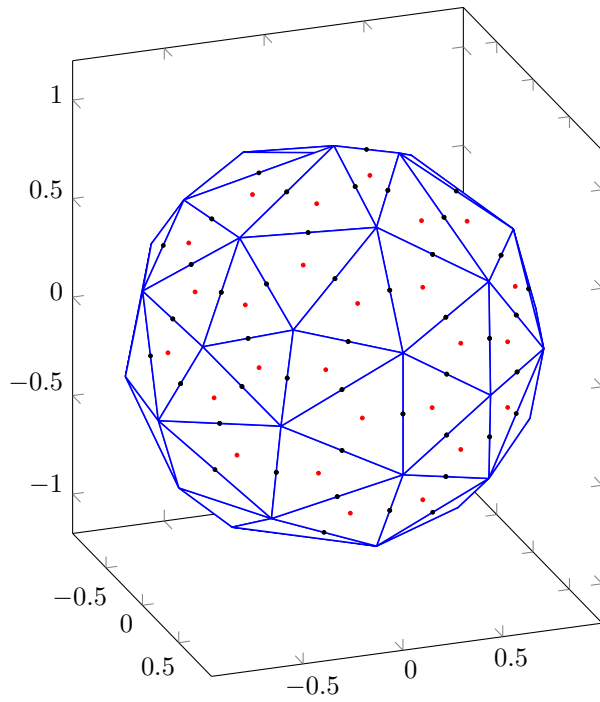
Main assumption. We assume either that all the primal control volumes $\mathcal{K} \in \mathfrak{M}$ are star-shaped with respect to $x_{\mathcal{K}}$ either that all the dual control volumes $\mathcal{K}^* \in \mathfrak{M}^*$ are star-shaped with respect to $x_{\mathcal{K}^*}$.

Remarque I.1.5. This hypothesis is not so restrictive and is fulfilled for instance in the case of a primal Voronoï mesh associated to the vertices $x_{\mathcal{K}^*}$, or in the dual case of a Delaunay conformal triangulation of the domain.

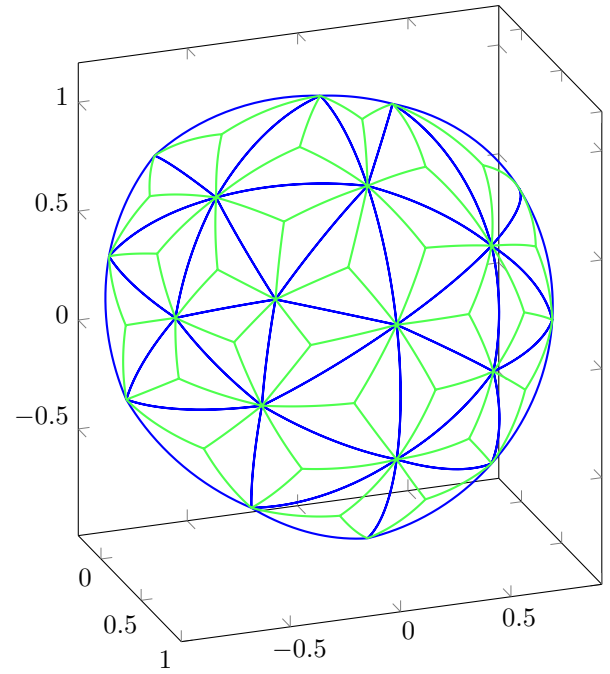
I.1.5 Notations

For any control volume $\mathcal{K} \in \mathfrak{M}$, we define

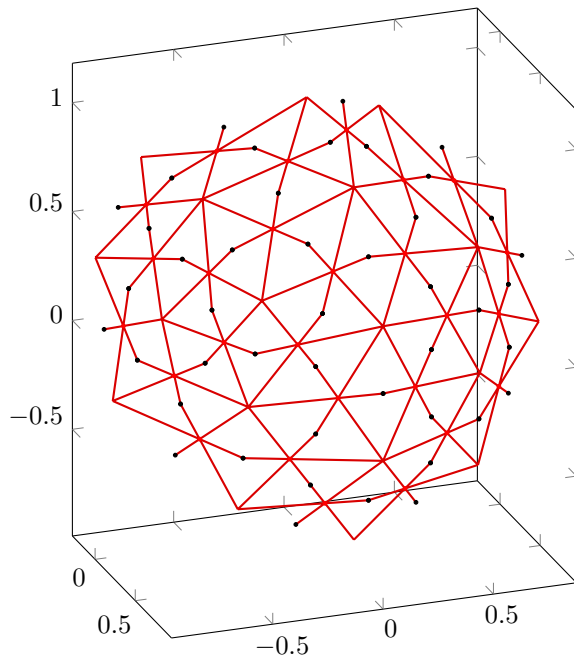
- $m(\mathcal{K})$, the measure of $\mathcal{K}$;
- $\mathfrak{E}_{\mathcal{K}}$, the set of edges of $\mathcal{K}$;
- $\mathcal{D}_{\mathcal{K}} = \{\mathcal{D}_{\sigma, \sigma^*} \in \mathcal{D} \mid \sigma \in \mathfrak{E}_{\mathcal{K}}\}$;



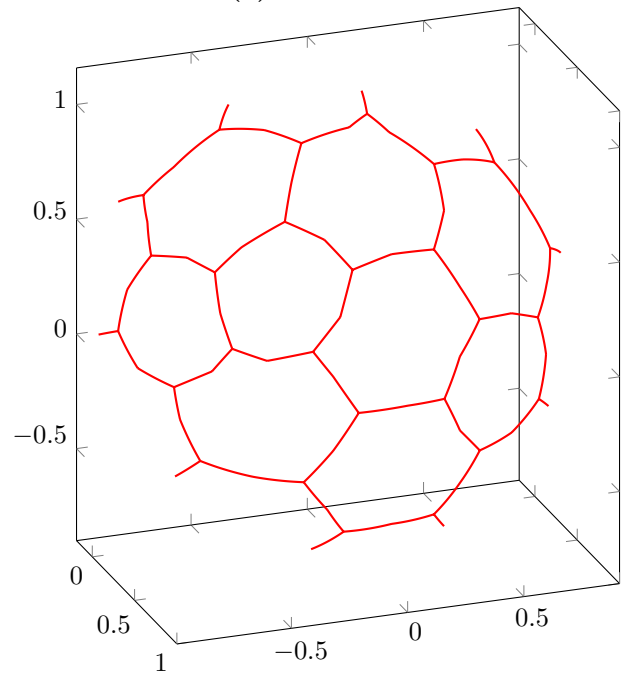
(a) Primal polyédrique



(b) Primal lisse



(c) Dual polyédrique



(d) Dual lisse

FIGURE I.6

- $\nu_{\mathcal{K}}$, the outward unit normal vector to $\partial\mathcal{K}$;
- $d_{\mathcal{K}}$, the diameter of $\mathcal{K}$.

In the same way, for a dual control volume $\mathcal{K}^* \in \mathfrak{M}^*$, we set

- $m(\mathcal{K}^*)$, the measure of $\mathcal{K}^*$;
- $\mathfrak{E}_{\mathcal{K}^*}$, the set of edges of $\mathcal{K}^*$;
- $\mathcal{D}_{\mathcal{K}^*} = \{\mathcal{D}_{\sigma, \sigma^*} \in \mathcal{D} \mid \sigma^* \in \mathfrak{E}_{\mathcal{K}^*}\}$;
- $\nu_{\mathcal{K}^*}$, the outward unit normal vector to $\partial\mathcal{K}^*$;
- $d_{\mathcal{K}^*}$, the diameter of $\mathcal{K}^*$.

For a diamond cell $\mathcal{D}_{\sigma, \sigma^*}$, recall that $(x_{\mathcal{K}}, x_{\mathcal{K}^*}, x_{\mathcal{L}}, x_{\mathcal{L}^*})$ are the vertices of $\mathcal{D}_{\sigma, \sigma^*}$ and note (I.2b)

- $m(\sigma)$, the length of σ , $m(\sigma^*)$ the length of σ^* and $m(\mathcal{D})$ the measure of the diamond cell;
- τ , the unit vector parallel to σ , oriented from $x_{\mathcal{K}^*}$ to $x_{\mathcal{L}^*}$;
- ν , the unit vector normal to σ , oriented from $x_{\mathcal{K}}$ to $x_{\mathcal{L}}$;
- τ^* , the unit vector parallel to σ^* , oriented from $x_{\mathcal{K}}$ to $x_{\mathcal{L}}$;
- ν^* , the unit vector normal to σ^* , oriented from $x_{\mathcal{K}^*}$ to $x_{\mathcal{L}^*}$;
- $\alpha_{\mathcal{D}}$, the angle between τ and τ^* ;
- $d_{\mathcal{D}}$, the diameter of $\mathcal{D}_{\sigma, \sigma^*}$.

It can happen that the point $x_{\mathcal{K}}$ does not belong to $\mathcal{K}$ (if $\mathcal{K}$ is a triangle and $x_{\mathcal{K}}$ its circumcenter for instance). As usual in that case (see [EGH00]) it is necessary to assume that $\nu_{\mathcal{K}}$ points from $x_{\mathcal{K}}$ towards $x_{\mathcal{L}}$, that is $\nu_{\mathcal{K}} = \nu$ with the notations above. Similarly we assume that $\nu_{\mathcal{K}^*} = \nu^*$. In the case of a triangular mesh $\mathfrak{M}$ in which $x_{\mathcal{K}}$ is the circumcenter of $\mathcal{K}$, this assumption is known as the Delaunay condition.

I.1.6 Unknowns

The finite volume method associates to all primal control volumes $\mathcal{K} \in \mathfrak{M}$, an unknown value $u_{\mathcal{K}}$ and to all dual control volumes $\mathcal{K}^* \in \mathfrak{M}^*$, an unknown value $u_{\mathcal{K}^*}$. We denote the approximate solution on the mesh $\mathcal{T}$ by

$$u^{\mathcal{T}} = ((u_{\mathcal{K}})_{\mathcal{K} \in \mathfrak{M}}, (u_{\mathcal{K}^*})_{\mathcal{K}^* \in \mathfrak{M}^*}). \quad (\text{I.1.1})$$

The space of all discrete functions $u^{\mathcal{T}}$ in the sense of definition I.1.1 is denoted by $\mathbb{R}^{\mathcal{T}}$. Whenever it is convenient, we associate to the discrete function $u^{\mathcal{T}} = ((u_{\mathcal{K}})_{\mathcal{K} \in \mathfrak{M}}, (u_{\mathcal{K}^*})_{\mathcal{K}^* \in \mathfrak{M}^*})$ the piecewise constant function

$$u^{\mathcal{T}} \sim \frac{1}{2}(u^{\mathfrak{M}} + u^{\mathfrak{M}^*}),$$

where $u^{\mathfrak{M}} \stackrel{\text{def.}}{=} \sum_{\mathcal{K} \in \mathfrak{M}} \mathbf{1}_{\mathcal{K}} u_{\mathcal{K}}$ and $u^{\mathfrak{M}^*} \stackrel{\text{def.}}{=} \sum_{\mathcal{K}^* \in \mathfrak{M}^*} \mathbf{1}_{\mathcal{K}^*} u_{\mathcal{K}^*}$. As a consequence, one can define for any $r \in [1; +\infty]$ the L^r norm of $u^{\mathfrak{M}}, u^{\mathfrak{M}^*}, u^{\mathcal{T}}$. We denote by $\llbracket \cdot, \cdot \rrbracket_{\mathcal{T}}$ the inner product on $\mathbb{R}^{\mathcal{T}}$ given by

$$\llbracket u^{\mathcal{T}}, v^{\mathcal{T}} \rrbracket_{\mathcal{T}} = \frac{1}{2} \sum_{\mathcal{K} \in \mathfrak{M}} m(\mathcal{K}) u_{\mathcal{K}} v_{\mathcal{K}} + \frac{1}{2} \sum_{\mathcal{K}^* \in \mathfrak{M}^*} m(\mathcal{K}^*) u_{\mathcal{K}^*} v_{\mathcal{K}^*},$$

which stands for a discrete L^2 inner product. Whereas the usual Euclidean inner product on $\mathbb{R}^{\mathcal{T}}$ is denoted by $(\cdot, \cdot)$:

$$(u^{\mathcal{T}}, v^{\mathcal{T}}) = \sum_{\mathcal{K} \in \mathfrak{M}} u_{\mathcal{K}} v_{\mathcal{K}} + \sum_{\mathcal{K}^* \in \mathfrak{M}^*} u_{\mathcal{K}^*} v_{\mathcal{K}^*}.$$

$$\forall \xi^{\mathcal{D}}, \phi^{\mathcal{D}} \in (\mathbb{R}^2)^{\mathcal{D}}, \quad (\xi^{\mathcal{D}}, \phi^{\mathcal{D}})_{\mathcal{D}} = \sum_{\mathcal{D} \in \mathcal{D}} m(\mathcal{D}) \xi^{\mathcal{D}} \cdot \phi^{\mathcal{D}}.$$

$$\|u^{\mathcal{T}}\|_2 \stackrel{\text{def.}}{=} \llbracket u^{\mathcal{T}}, u^{\mathcal{T}} \rrbracket_{\mathcal{T}}^{1/2}, \quad \forall u^{\mathcal{T}} \in \mathbb{R}^{\mathcal{T}}$$

$$\|\xi^{\mathcal{D}}\|_2 \stackrel{\text{def.}}{=} (\xi^{\mathcal{D}}, \xi^{\mathcal{D}})_{\mathcal{D}}^{1/2}, \quad \forall \xi^{\mathcal{D}} \in (\mathbb{R}^2)^{\mathcal{D}}$$

I.1.7 Discrete gradient [ABH07; Kre10]

We consider the discrete gradient introduced by Coudière and *al.* in [CVV99] and applied by Hermeline [Her00] and Domelevo and Omnès [DO05] in the framework of DDFV schemes described above.

Définition I.1.6 (Discrete gradient). The *discrete gradient operator* $\nabla^{\mathcal{D}} : \mathbb{R}^T \rightarrow (\mathbb{R}^2)^{\mathcal{D}}$ can be defined as follows: for any $u^T \in \mathbb{R}^T$, $\nabla^{\mathcal{D}} u^T = (\nabla^{\mathcal{D}} u^T)_{\mathcal{D} \in \mathcal{D}}$ is the function, constant on each diamond cell $\mathcal{D}_{\sigma, \sigma^*}$, given by

$$\nabla^{\mathcal{D}} u^T = \sum_{\mathcal{D} \in \mathcal{D}} \nabla^{\mathcal{D}} u^T \chi_{\mathcal{D}},$$

with

$$(\nabla^{\mathcal{D}} u^T, \tau) = \frac{u_{\mathcal{L}^*} - u_{\mathcal{K}^*}}{m(\sigma)}, \quad (\nabla^{\mathcal{D}} u^T, \tau^*) = \frac{u_{\mathcal{L}} - u_{\mathcal{K}}}{m(\sigma^*)},$$

where $\mathcal{D}_{\sigma, \sigma^*}$ is noted $\mathcal{D}$ when no confusion can arise and $\chi_{\mathcal{D}}$ is defined by I.1.4.

Remark that $\nabla^{\mathcal{D}} u^T$ can be expressed in the (ν, ν^*) basis in the following way:

$$\nabla^{\mathcal{D}} u^T = \frac{1}{\sin \alpha_{\mathcal{D}}} \left(\frac{u_{\mathcal{L}} - u_{\mathcal{K}}}{m(\sigma^*)} \nu + \frac{u_{\mathcal{L}^*} - u_{\mathcal{K}^*}}{m(\sigma)} \nu^* \right)$$

thanks to the following lemma.

Lemme I.1.7. Using the notations and orientation conventions of **à compléter**, for any vector $\xi \in \mathbb{R}^2$ we have

$$(\sin \alpha_{\mathcal{D}}) \xi = (\xi, \tau) \nu^* + (\xi, \tau^*) \nu.$$

L'aire $m(\mathcal{D})$ d'un diamant $\mathcal{D}$ est la suivante : $m(\mathcal{D}) = \frac{1}{2} m(\sigma) m(\sigma^*) \sin(\alpha_{\mathcal{D}})$, on peut alors réécrire le gradient discret:

$$\nabla^{\mathcal{D}} u^T = \frac{1}{2m(\mathcal{D})} \left[(u_{\mathcal{L}} - u_{\mathcal{K}}) m(\sigma) \nu + (u_{\mathcal{L}^*} - u_{\mathcal{K}^*}) m(\sigma^*) \nu^* \right].$$

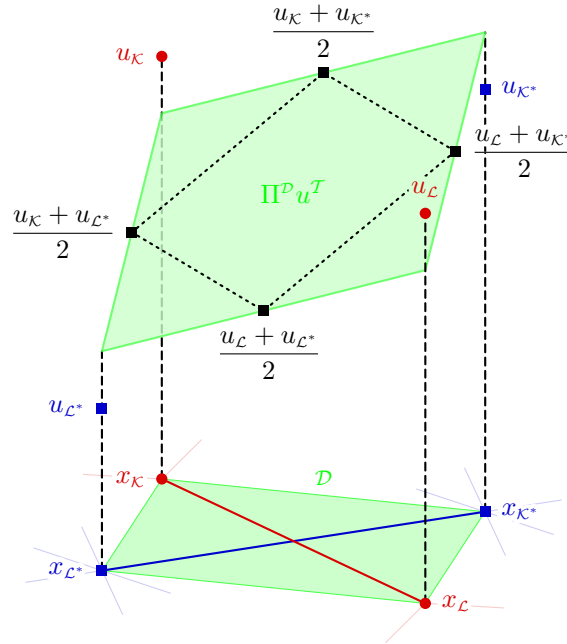


FIGURE I.8. Projection DDFV

I.1.8 Discrete vector curl operator

Définition I.1.8 (Discrete vector curl operator). The *discrete vector curl operator* $\mathbf{rot}^{\mathcal{D}} : \mathbb{R}^T \rightarrow (\mathbb{R}^2)^{\mathcal{D}}$ can be defined as follows: for any $u^T \in \mathbb{R}^T$, $\mathbf{rot}^{\mathcal{D}} u^T = (\mathbf{rot}^{\mathcal{D}} u^T)_{\mathcal{D} \in \mathcal{D}}$ is the function, constant on each diamond cell $\mathcal{D}_{\sigma, \sigma^*}$, given

by

$$\mathbf{rot}^{\mathcal{D}} u^{\mathcal{T}} = -\frac{1}{2m(\mathcal{D})} \left[(u_{\mathcal{L}} - u_{\mathcal{K}}) m(\sigma) \boldsymbol{\tau} + (u_{\mathcal{L}^*} - u_{\mathcal{K}^*}) m(\sigma^*) \boldsymbol{\tau}^* \right].$$

I.1.9 Discrete divergence

Définition I.1.9 (Discrete divergence). L'opérateur de divergence discrète est défini de la manière suivante: $\text{div}^{\mathcal{T}} : (\mathbb{R}^2)^{\mathcal{D}} \rightarrow \mathbb{R}^{\mathcal{T}}$. Soit $\xi^{\mathcal{D}} \in (\mathbb{R}^2)^{\mathcal{D}}$, on pose

$$\text{div}^{\mathcal{T}} \xi^{\mathcal{D}} = (\text{div}^{\mathfrak{M}} \xi^{\mathcal{D}}, \text{div}^{\mathfrak{M}^*} \xi^{\mathcal{D}}),$$

avec $\text{div}^{\mathfrak{M}} \xi^{\mathcal{D}} = (\text{div}^{\mathcal{K}} \xi^{\mathcal{D}})_{\mathcal{K} \in \mathfrak{M}}$, $\text{div}^{\mathfrak{M}^*} \xi^{\mathcal{D}} = (\text{div}^{\mathcal{K}^*} \xi^{\mathcal{D}})_{\mathcal{K}^* \in \mathfrak{M}^*}$:

$$\forall \mathcal{K} \in \mathfrak{M}, \quad \text{div}^{\mathcal{K}} \xi^{\mathcal{D}} = \frac{1}{m(\mathcal{K})} \sum_{\mathcal{D}_{\sigma, \sigma^*} \in \mathcal{D}_{\mathcal{K}}} m(\sigma) \xi^{\mathcal{D}} \cdot \boldsymbol{\nu},$$

$$\forall \mathcal{K}^* \in \mathfrak{M}^*, \quad \text{div}^{\mathcal{K}^*} \xi^{\mathcal{D}} = \frac{1}{m(\mathcal{K}^*)} \sum_{\mathcal{D}_{\sigma, \sigma^*} \in \mathcal{D}_{\mathcal{K}^*}} m(\sigma^*) \xi^{\mathcal{D}} \cdot \boldsymbol{\nu}^*.$$

I.1.10 Discrete scalar curl operator

Définition I.1.10 (Discrete scalar curl operator). $\text{rot}^{\mathcal{T}} : (\mathbb{R}^2)^{\mathcal{D}} \rightarrow \mathbb{R}^{\mathcal{T}}$. $\xi^{\mathcal{D}} \in (\mathbb{R}^2)^{\mathcal{D}}$

$$\text{rot}^{\mathcal{T}} \xi^{\mathcal{D}} = (\text{rot}^{\mathfrak{M}} \xi^{\mathcal{D}}, \text{rot}^{\mathfrak{M}^*} \xi^{\mathcal{D}}),$$

with $\text{rot}^{\mathfrak{M}} \xi^{\mathcal{D}} = (\text{rot}^{\mathcal{K}} \xi^{\mathcal{D}})_{\mathcal{K} \in \mathfrak{M}}$, $\text{rot}^{\mathfrak{M}^*} \xi^{\mathcal{D}} = (\text{rot}^{\mathcal{K}^*} \xi^{\mathcal{D}})_{\mathcal{K}^* \in \mathfrak{M}^*}$:

$$\forall \mathcal{K} \in \mathfrak{M}, \quad \text{rot}^{\mathcal{K}} \xi^{\mathcal{D}} = \frac{1}{m(\mathcal{K})} \sum_{\mathcal{D}_{\sigma, \sigma^*} \in \mathcal{D}_{\mathcal{K}}} m(\sigma) \xi^{\mathcal{D}} \cdot \boldsymbol{\tau},$$

$$\forall \mathcal{K}^* \in \mathfrak{M}^*, \quad \text{rot}^{\mathcal{K}^*} \xi^{\mathcal{D}} = \frac{1}{m(\mathcal{K}^*)} \sum_{\mathcal{D}_{\sigma, \sigma^*} \in \mathcal{D}_{\mathcal{K}^*}} m(\sigma^*) \xi^{\mathcal{D}} \cdot \boldsymbol{\tau}^*.$$

I.1.11 Discrete Green formulae

Here, we check that the discrete operators verify some discrete duality principles.

Théorème I.1.11 (Green formula). Pour tout $\xi^{\mathcal{D}}, u^{\mathcal{T}} \in (\mathbb{R}^2)^{\mathcal{D}} \times \mathbb{R}^{\mathcal{T}}$, on a

$$\llbracket \text{div}^{\mathcal{T}} \xi^{\mathcal{D}}, u^{\mathcal{T}} \rrbracket_{\mathcal{T}} = -(\xi^{\mathcal{D}}, \nabla^{\mathcal{D}} u^{\mathcal{T}})_{\mathcal{D}},$$

$$\llbracket \text{rot}^{\mathcal{T}} \xi^{\mathcal{D}}, u^{\mathcal{T}} \rrbracket_{\mathcal{T}} = (\xi^{\mathcal{D}}, \mathbf{rot}^{\mathcal{D}} u^{\mathcal{T}})_{\mathcal{D}}.$$

■ Démonstration.

□

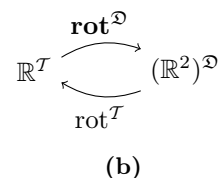
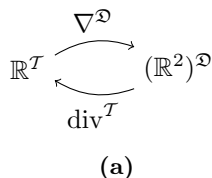


FIGURE I.9

I.1.12 Regularity of meshes

- On note $\sigma_{\mathcal{K}} \stackrel{\text{déf.}}{=} B(x_{\mathcal{K}}, \rho_{\mathcal{K}})$ et $\sigma_{\mathcal{K}^*} \stackrel{\text{déf.}}{=} B(x_{\mathcal{K}^*}, \rho_{\mathcal{K}^*})$ and $\rho_{\mathcal{K}}$ and $\rho_{\mathcal{K}^*}$ are positive numbers associated to the mesh $\mathcal{T}$ and such that

$$\sigma_{\mathcal{K}} \subset \bar{\mathcal{K}}, \quad \sigma_{\mathcal{K}^*} \subset \partial \mathcal{K}^*.$$

Finally, introduce numbers $\rho_{\mathcal{K}}$ and $\rho_{\mathcal{K}^*}$ for any $\mathcal{K} \in \mathfrak{M}$ (resp. $\mathcal{K}^* \in \mathfrak{M}^*$) such that

$$B_{\mathcal{K}} \stackrel{\text{déf.}}{=} B(x_{\mathcal{K}}, \rho_{\mathcal{K}}) \subset \Omega, \quad B_{\mathcal{K}^*} \stackrel{\text{déf.}}{=} B(x_{\mathcal{K}^*}, \rho_{\mathcal{K}^*}) \subset \Omega.$$

These balls are only introduced in order to prove the convergence of the scheme.

We note $\text{size}(\mathcal{T})$ the maximum of the diameters of the diamond cells in $\mathfrak{D}$:

$$\text{size}(\mathcal{T}) \stackrel{\text{déf.}}{=} \max_{\mathcal{D} \in \mathfrak{D}} d_{\mathcal{D}}.$$

The following bounds follow:

$$m(\sigma) \leq \text{size}(\mathcal{T}), \quad \forall \sigma \in \mathfrak{E}; \quad m(\sigma^*) \leq \text{size}(\mathcal{T}), \quad \forall \sigma^* \in \mathfrak{E}^*;$$

$$m(\mathcal{K}) \leq \pi \text{size}(\mathcal{T})^2, \quad \forall \mathcal{K} \in \mathfrak{M}; \quad m(\mathcal{K}^*) \leq \pi \text{size}(\mathcal{T})^2, \quad \forall \mathcal{K}^* \in \mathfrak{M}^*; \quad m(\mathcal{D}) \leq \frac{1}{2} \text{size}(\mathcal{T})^2, \quad \forall \mathcal{D} \in \mathfrak{D}.$$

We introduce now a positive number that quantifies the regularity of a given mesh and is useful to perform the convergence analysis of the finite volume schemes. We first define $\alpha_{\mathcal{T}}$ to be the unique real number in $]0; \frac{\pi}{2}]$ such that

$$\sin \alpha_{\mathcal{T}} \stackrel{\text{déf.}}{=} \min_{\mathcal{D} \in \mathfrak{D}} |\sin \alpha_{\mathcal{D}}|,$$

that is the minimal angle between the diagonals of the diamond cells in the mesh.

Let us introduce the number

$$\begin{aligned} \mathcal{N}_{\mathcal{T}} \stackrel{\text{déf.}}{=} & \sup_{x \in \Omega} \# \left\{ \mathcal{K}^* \mid x \in \widehat{\mathcal{K}^* \cup B_{\mathcal{K}^*}} \right\} + \sup_{x \in \Omega} \# \left\{ \mathcal{K} \mid x \in \widehat{\mathcal{K} \cup B_{\mathcal{K}}} \right\} \\ & + \sup_{x \in \Omega} \# \left\{ \mathcal{D} \mid x \in \widehat{\mathcal{D} \cup B_{\mathcal{K}}}, \mathcal{D} \in \mathfrak{D}_{\mathcal{K}}, \mathcal{K} \in \mathfrak{M} \right\} \\ & + \sup_{x \in \Omega} \# \left\{ \mathcal{D} \mid x \in \widehat{\mathcal{D} \cup B_{\mathcal{K}^*}}, \mathcal{D} \in \mathfrak{D}_{\mathcal{K}^*}, \mathcal{K}^* \in \mathfrak{M}^* \right\}, \end{aligned}$$

where $\widehat{E}$ denotes the convex hull of any set $E \subset \mathbb{R}^2$. In the usual case where the primal control volumes, the dual control volumes and the diamond cells are **convex** and where $B_{\mathcal{K}} \subset \mathcal{K}$, $B_{\mathcal{K}^*} \subset \mathcal{K}^*$ then $\mathcal{N}_{\mathcal{T}}$ can be bounded by a function of the maximal number of edges per primal and dual control volumes. Since we do not impose any convexity assumption on the meshes, we need to control this number $\mathcal{N}_{\mathcal{T}}$ in the convergence analysis. We can now introduce

$$\begin{aligned} \text{reg } \mathcal{T} \stackrel{\text{déf.}}{=} & \max \left\{ \frac{1}{\alpha_{\mathcal{T}}}, \mathcal{N}_{\mathcal{T}}, \max_{\mathcal{D} \in \mathfrak{D}} \frac{d_{\mathcal{D}}}{\sqrt{m(\mathcal{D})}}, \max_{\mathcal{K} \in \mathfrak{M}} \frac{d_{\mathcal{K}}}{\sqrt{m(\mathcal{K})}}, \max_{\mathcal{K}^* \in \mathfrak{M}^*} \frac{d_{\mathcal{K}^*}}{\sqrt{m(\mathcal{K}^*)}}, \right. \\ & \left. \max_{\mathcal{K} \in \mathfrak{M}} \left(\frac{d_{\mathcal{K}}}{\rho_{\mathcal{K}}} + \frac{\rho_{\mathcal{K}}}{d_{\mathcal{K}}} \right), \max_{\mathcal{K}^* \in \mathfrak{M}^*} \left(\frac{d_{\mathcal{K}^*}}{\rho_{\mathcal{K}^*}} + \frac{\rho_{\mathcal{K}^*}}{d_{\mathcal{K}^*}} \right), \max_{\substack{\mathcal{K} \in \mathfrak{M} \\ \mathcal{D} \in \mathfrak{D}_{\mathcal{K}}}} \frac{d_{\mathcal{K}}}{d_{\mathcal{D}}}, \max_{\substack{\mathcal{K}^* \in \mathfrak{M}^* \\ \mathcal{D} \in \mathfrak{D}_{\mathcal{K}^*}}} \frac{d_{\mathcal{K}^*}}{d_{\mathcal{D}}} \right\}. \end{aligned}$$

Let us point out that $\text{reg}(\mathcal{T})$ essentially measures:

- how flat the diamond cells are;
- how large is the difference between the size of a primal control volume (resp. a dual control volume) and the size of a diamond cell as soon as they intersect.

Convention. In any estimate given, the dependence of the constants in $\text{reg}(\mathcal{T})$ is implicitly assumed to be non-decreasing. Furthermore, the dependence of the constants on the domain is often omitted.

Remarque I.1.12. For conformal finite volume meshes, it is assumed that $\alpha_{\mathcal{D}} = \frac{\pi}{2}$ for any diamond $\mathcal{D} \in \mathfrak{D}$, so that $\alpha_{\mathcal{T}} = \frac{\pi}{2}$. In our case, not only this orthogonality condition is relaxed but also $\mathfrak{M}$ can present atypical edges, non convex dual control volumes and non convex diamond cells [figure](#).

Remarque I.1.13. The boundedness of $\text{reg}(\mathcal{T})$ imposes only local restriction on the mesh. It is easy to construct a family of locally refined mesh such that $\text{reg}(\mathcal{T})$ is bounded independently on the level of the refinement.

I.1.13 Régularité du maillage [Kre10, p. 36]

Soit $\text{size}(\mathcal{T})$ le maximum des diamètres des diamants. Pour mesurer l'applatissage des diamants, on note $\alpha_{\mathcal{T}}$ l'unique réel dans $]0; \frac{\pi}{2}]$ tel que $\sin(\alpha_{\mathcal{T}}) \stackrel{\text{déf.}}{=} \min_{\mathcal{D} \in \mathfrak{D}} |\sin(\alpha_{\mathcal{D}})|$ si la maillage dual est **direct** et $\sin(\alpha_{\mathcal{T}}) \stackrel{\text{déf.}}{=} \min_{\mathcal{D} \in \mathfrak{D}} (|\sin(\alpha_{\mathcal{K}})|, |\sin(\alpha_{\mathcal{L}})|)$ si le maillage dual est **barycentrique**. On introduit un nombre positif $\text{reg}(\mathcal{T})$ qui mesure la régularité d'un maillage donné et qui est utilisé pour réaliser la convergence des schémas:

$$\mathcal{N}_{\mathcal{T}} = \sup_{x \in \Omega} \# \left\{ \mathcal{D} \mid x \in \widehat{\mathcal{D} \cup \mathcal{K}}, \mathcal{D} \in \mathfrak{D}_{\mathcal{K}}, \mathcal{K} \in \mathfrak{M} \right\} + \sup_{x \in \Omega} \# \left\{ \mathcal{D} \mid x \in \widehat{\mathcal{D} \cup \mathcal{K}^*}, \mathcal{D} \in \mathfrak{D}_{\mathcal{K}^*}, \mathcal{K}^* \in \mathfrak{M}^* \right\},$$

$$\text{reg}(\mathcal{T}) = \max \left\{ \frac{1}{\sin(\alpha_{\mathcal{T}})}, \mathcal{N}, \mathcal{N}^*, \mathcal{N}_{\mathcal{T}}, \max_{\mathcal{D} \in \mathfrak{D}} \max_{\mathcal{Q} \in \Omega_{\mathcal{D}}} \frac{d_{\mathcal{D}}}{\min_{\sigma \in \partial \mathcal{Q}} m(\sigma)}, \max_{\mathcal{K} \in \mathfrak{M}} \frac{d_{\mathcal{K}}}{\sqrt{m(\mathcal{K})}}, \max_{\mathcal{K}^* \in \mathfrak{M}^*} \frac{d_{\mathcal{K}^*}}{\sqrt{m(\mathcal{K}^*)}}, \right. \\ \left. \max_{\mathcal{K} \in \mathfrak{M}} \max_{\mathcal{D} \in \mathfrak{D}_{\mathcal{K}}} \frac{d_{\mathcal{K}}}{d_{\mathcal{D}}}, \max_{\mathcal{K}^* \in \mathfrak{M}^*} \max_{\mathcal{D} \in \mathfrak{D}_{\mathcal{K}^*}} \frac{d_{\mathcal{K}^*}}{d_{\mathcal{D}}} \right\},$$

où $\widehat{P}$ désigne l'enveloppe convexe d'un ensemble P , $\mathcal{N}$, $\mathcal{N}^*$ sont le nombre maximum d'arêtes d'une maille primale et le nombre maximum d'arêtes incidentes d'un sommet du maillage primal. Ce nombre $\text{reg}(\mathcal{T})$ mesure essentiellement l'applatissage des diamants et le rapport entre la taille des mailles primales (resp. duales) et la taille des diamants et il doit être uniformément borné quand $\text{size}(\mathcal{T}) \rightarrow 0$ pour avoir la convergence. Par exemple, ce nombre $\text{reg}(\mathcal{T})$ implique les résultats géométriques suivants: il existe une constante $C > 0$, dépendant uniquement de $\text{reg}(\mathcal{T})$ telle que

$$\frac{d_{\mathcal{D}}}{\sqrt{m(\mathcal{D})}} \leq C, \forall \mathcal{D} \in \mathfrak{D}, \quad \frac{d_{\mathcal{Q}}}{\sqrt{m(\mathcal{Q})}} \leq C, \forall \mathcal{Q} \in \Omega \quad \text{et} \quad d_{\mathcal{D}} \leq C \min(m(\sigma), m(\sigma^*)), \forall \mathcal{D}_{\sigma, \sigma^*} \in \mathfrak{D}.$$

Hypothèses. On suppose que toute maille primale $\mathcal{K} \in \mathfrak{M}$ est étoilée par rapport à $x_{\mathcal{K}}$.

On suppose que toute maille duale $\mathcal{K}^* \in \mathfrak{M}^*$ est étoilée par rapport à $x_{\mathcal{K}^*}$.

Remarque I.1.14. On a que toute maille diamant $\mathcal{D}$ est étoilée par rapport à $x_{\mathcal{D}}$.

Remarque I.1.15. Grâce aux hypothèses ci-dessus, on a également:

$$\text{diam}(\widehat{K}) \leq 2 d_{\mathcal{K}}, \forall \mathcal{K} \in \mathfrak{M} \quad \text{et} \quad \text{diam}(\widehat{K}^*) \leq 2 d_{\mathcal{K}^*}, \forall \mathcal{K}^* \in \mathfrak{M}^*,$$

de plus la remarque [I.1.14](#) implique:

$$\text{diam}(\widehat{\mathcal{D}}) \leq 2 d_{\mathcal{D}}, \forall \mathcal{D} \in \mathfrak{D}.$$

I.2 Projection

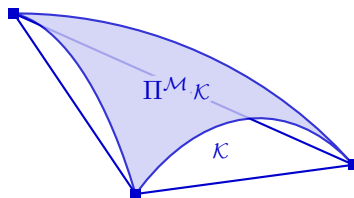


FIGURE I.10. A control volume $\mathcal{K}$ and its projection $\Pi^{\mathcal{M}} \mathcal{K}$ onto the surface $\mathcal{M}$

I.2.1 Global coordinates

[DE13, Sec. 2.3] Assume in the following that $G \subset \mathbb{R}^{n+1}$ is bounded and open with exterior normal ν and assume that $\mathcal{M} = \partial G$ is a $\mathcal{C}^k$ -hypersurface ($k \geq 2$). The *oriented distance function* for $\mathcal{M}$ is defined by

$$d(x) \stackrel{\text{d\'ef.}}{=} \begin{cases} \inf_{y \in \mathcal{M}} |x - y| & x \in \mathbb{R}^{n+1} \setminus \overline{G} \\ - \inf_{y \in \mathcal{M}} |x - y| & x \in G \end{cases}.$$

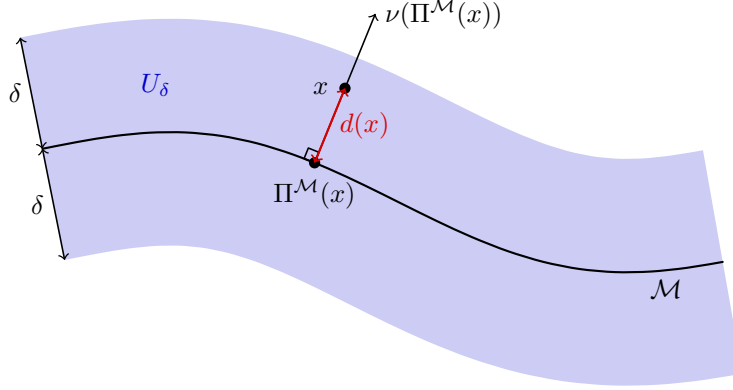


FIGURE I.11. Strip U_δ around the hypersurface $\mathcal{M}$ and normal coordinates $x = \Pi^\mathcal{M}(x) + d(x)\nu(\Pi^\mathcal{M}(x))$

Lemme I.2.1 ([DE13, Lemma 2.8]). We define

$$U_\delta \stackrel{\text{d\'ef.}}{=} \{x \in \mathbb{R}^{n+1} \mid |d(x)| < \delta\}.$$

Then $d \in \mathcal{C}^k(U_\delta)$, and for every point $x \in U_\delta$ there exists a unique point $\Pi^\mathcal{M}(x) \in \mathcal{M}$ such that

$$x = \Pi^\mathcal{M}(x) + d(x)\nu(\Pi^\mathcal{M}(x)). \quad (\text{I.2.1})$$

Moreover, we have that

$$\nabla d(x) = \nu(\Pi^\mathcal{M}(x)), \quad |\nabla d(x)| = 1, \quad \forall x \in U_\delta.$$

We also extend the normal constantly in the normal direction: $\nu(x) = \nu(\Pi^\mathcal{M}(x))$ for $x \in U_\delta$. Thus we have introduced a global coordinate system around $\mathcal{M}$. Every point $x \in U_\delta$ can be described by its Fermi coordinates (normal coordinates) $d(x)$ and $\Pi^\mathcal{M}(x)$ according to I.2.1.

À mettre dans régularité du maillage. [DE13, Sec. 4.1] We assume that the surface $\mathcal{M}_h$ which approximates the smooth n -dimensional surface $\mathcal{M}$ ($\partial\mathcal{M} = \emptyset$) lies in the strip U_δ (see Lemma I.2.1) and is a Lipschitz surface. The strip can be chosen with locally varying width: see I.2.2. We assume that the discrete triangulated surface $\mathcal{M}_h$ besides I.2.2 has the following properties. The surface $\mathcal{M}_h$ is the union of finitely many non-degenerate (closed) n -simplices in $\mathbb{R}^{n+1}$. We let $\mathfrak{M}$ denote the set of these simplices:

$$\mathcal{M}_h = \bigcup_{\kappa \in \mathfrak{M}} \kappa.$$

The vertices $\mathfrak{B}^*$ of the simplices are taken to sit on the smooth surface $\mathcal{M}$. For $\kappa, \mathcal{L} \in \mathfrak{M}$, either $\kappa \cap \mathcal{L} = \emptyset$ or $\kappa \cap \mathcal{L}$ is an $(n - k)$ -dimensional side simplex ($k \in \{1, \dots, n\}$) common to both of the simplices κ and $\mathcal{L}$. For $\kappa \in \mathfrak{M}$ we denote by d_κ its diameter and by $r(\kappa)$ the in-ball radius. Also,

$$h \stackrel{\text{d\'ef.}}{=} \max_{\kappa \in \mathfrak{M}} d_\kappa, \quad r \stackrel{\text{d\'ef.}}{=} \min_{\kappa \in \mathfrak{M}} r(\kappa).$$

We assume that the quantity

$$\sigma \stackrel{\text{d\'ef.}}{=} \max_{\kappa \in \mathfrak{M}} \sigma(\kappa), \quad \sigma(\kappa) = \frac{h(\kappa)}{r(\kappa)}$$

is uniformly bounded independently of h .

Note that, by Lemma I.2.1, for every simplex $\kappa \subset \mathcal{M}_h$ there is a unique curved simplex $\tilde{\kappa} = \Pi^{\mathcal{M}} \kappa \subset \mathcal{M}$. In order to avoid global double covering, we assume that for each point $\Pi^{\mathcal{M}} \in \mathcal{M}$ there is at most one point $x \in \mathcal{M}_h$ with $\Pi^{\mathcal{M}} = \Pi^{\mathcal{M}}(x)$. This implies that there is a bijective correspondence between the triangles on $\mathcal{M}_h$ and the induced curvilinear triangles on $\mathcal{M}$.

We assume that $\mathcal{M}$ has only one connected component. We start with a discrete surface $\mathcal{M}_h$ contained in the variable strip U_δ ,

$$\mathcal{M}_h \subset U_\delta = \{y + s\nu(y) \mid |s| < \delta(y), y \in \mathcal{M}\}, \quad (\text{I.2.2})$$

where for a point $y \in \mathcal{M}$ we set

$$\delta(y) = \min \left\{ \left(\max_{i=1,\dots,n} |\kappa_i(y)| \right)^{-1}, \sup_{z \in \mathcal{M}, z \neq y} \frac{L(y, z)}{|y - z|} \right\}$$

with the sectional curvatures κ_i and where $L(y, z)$ denotes the geodesic distance between y and z on $\mathcal{M}$ (see [Lee97] for the definitions of *sectional curvature* and *geodesic distance*).

Remarque I.2.2. We can take $\delta \stackrel{\text{d\'ef.}}{=} \min_{y \in \mathcal{M}} \delta(y)$ which exists since $\mathcal{M}$ is a compact manifold.

Proposition I.2.3. One can show that each point $x \in U_\delta$ can be uniquely projected onto the smooth surface, yielding the decomposition

$$x = \Pi^{\mathcal{M}}(x) + d(x)\nu(\Pi^{\mathcal{M}}(x)), \quad x \in U_\delta,$$

with $\Pi^{\mathcal{M}}(x)$ the orthogonal projection of x onto $\mathcal{M}$ and $d(x)$ the oriented distance between x and $\mathcal{M}$.

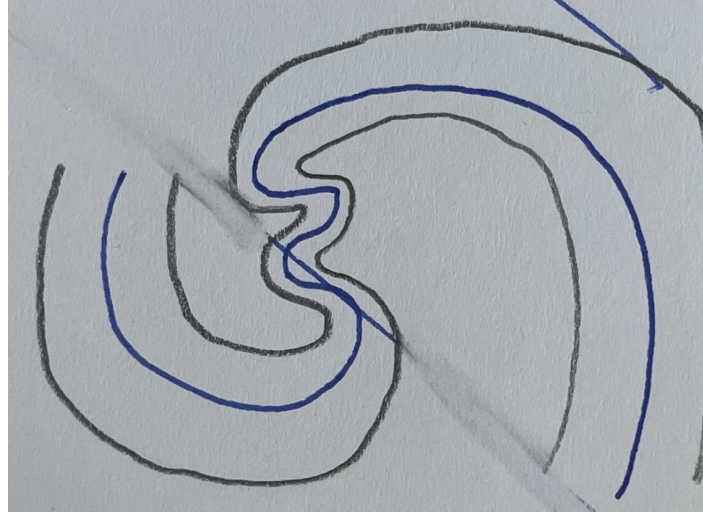


FIGURE I.12. $\delta(y)$

Proposition I.2.4.

$$\mathcal{M} = \bigcup_{\kappa \in \mathfrak{M}} \Pi^{\mathcal{M}} \kappa = \bigcup_{\kappa^* \in \mathfrak{M}^*} \Pi^{\mathcal{M}} \kappa^* = \bigcup_{\mathcal{D} \in \mathfrak{D}} \Pi^{\mathcal{M}} \mathcal{D}$$

I.2.2 Approximation of geometry [DE13, Sec. 4.2]

We consider a piecewise linear approximation of a smooth surface $\mathcal{M} \in \mathcal{C}^2$ and its geometry. We collect estimates for the geometric error which is produced by approximating the smooth surface $\mathcal{M}$ by a discrete surface $\mathcal{M}_h$.

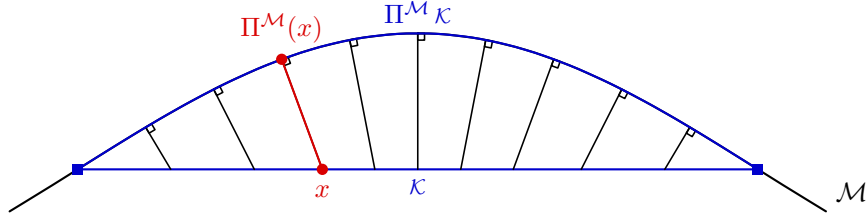


FIGURE I.13. A curved simplex $\Pi^{\mathcal{M}} \mathcal{K}$ is parametrized over a planar one. Orthogonal projection onto the smooth surface $\mathcal{M}$

Lemme I.2.5 (Projection [DE13, Lemma 4.1]). Assume $\mathcal{M}$ and $\mathcal{M}_h$ are as above. The map $\Pi^{\mathcal{M}}: \mathcal{M}_h \rightarrow \mathcal{M}$ is bijective. For the oriented distance function to $\mathcal{M}$ we have the estimate

$$\|d\|_{L^\infty(\mathcal{M}_h)} \leq c h^2.$$

The quotient, δ_h , between the smooth and discrete surface measures dA and dA_h , defined by $\delta_h dA_h = dA$, satisfies

$$\|1 - \delta_h\|_{L^\infty(\mathcal{M}_h)} \leq c h^2.$$

Let P and P_h be the projections onto the tangent planes, $P_{ij} = \delta_{ij} - \nu_i \nu_j$, $P_{h,ij} = \delta_{ij} - \nu_{h,i} \nu_{h,j}$, and let

$$R_h \stackrel{\text{def.}}{=} \frac{1}{\delta_h} P(I - d\mathcal{H}) P_h(I - d\mathcal{H}),$$

$\mathcal{H}_{ij} = d_{x_i x_j} = \nu_{i x_j}$. Then

$$\|(I - R_h)P\|_{L^\infty(\mathcal{M}_h)} \leq c h^2.$$

Lemme I.2.6.

$$\|d\|_{\mathfrak{M} + \mathfrak{M}^*} \leq c \text{size}(\mathcal{T})^2.$$

Lemme I.2.7. À plat, $m(\text{arête}) \sim h$ et $m(\text{triangle}) \sim h^2$. Sur la surface aussi.

For any function η defined on the discrete surface $\mathcal{M}_h$ we define an extension or lift onto $\mathcal{M}$ by

$$\eta^\ell(a) = \eta(x(a)), \quad a \in \mathcal{M}, \quad (\text{I.2.3})$$

where, by our assumptions, $x(a)$ is defined as the unique solution of

$$x = a + d(x) \nu(a).$$

Furthermore, we understand by $\eta^\ell(x)$ the constant extension from $\mathcal{M}$ in the normal direction $\nu(a(x))$.

In order to compare the norms between functions on $\mathcal{M}_h$ and their lift I.2.3 to $\mathcal{M}$ we need the following lemma.

Lemme I.2.8. Let $\eta: \mathcal{M}_h \rightarrow \mathbb{R}$ with the lift $\eta^\ell: \mathcal{M} \rightarrow \mathbb{R}$. Then, for the plane $T \subset \mathcal{M}_h$, and smooth curved triangles $\tilde{T} \subset \mathcal{M}$, the following estimates hold if the norm exist. There is a constant $c > 0$ independant of h such that

$$\begin{aligned} \frac{1}{c} \|\eta\|_{L^2(T)} &\leq \|\eta^\ell\|_{L^2(\tilde{T})} \leq c \|\eta\|_{L^2(T)}, \\ \frac{1}{c} \|\nabla_{\mathcal{M}_h} \eta\|_{L^2(T)} &\leq \|\nabla_{\mathcal{M}} \eta^\ell\|_{L^2(\tilde{T})} \leq c \|\nabla_{\mathcal{M}_h} \eta\|_{L^2(T)}, \\ \|\nabla_{\mathcal{M}_h}^2 \eta\|_{L^2(T)} &\leq c \|\nabla_{\mathcal{M}}^2 \eta^\ell\|_{L^2(\tilde{T})} + c h \|\nabla_{\mathcal{M}} \eta^\ell\|_{L^2(\tilde{T})}. \end{aligned}$$

I.2.3 Discrete charts [DE13, Sec. 4.3]

We add some information on a discrete differential geometric approach which is consistent with the introduction of parametrized surfaces. For a complete exposition see Nedelec (1976).

Let $\mathcal{M}$ be given as in Section 2.1 by local charts $(X_i)_{i \in I}$, $X_i \in \mathcal{C}^k(V_i, \mathbb{R}^{n+1})$. We assume that already

$$\mathcal{M} = \bigcup_{i \in I} X_i(V_{ih}),$$

with $V_{ih} \subset V_i$ and triangulated domains $V_{ih} \subset \mathbb{R}^n$. We interpolate the smooth parametrization $X_{ih} = I_h X_i$, where I_h is the interpolation from Lemma 4.3 on the polygonal domain V_{ih} . We assume that if $U_{ij} = X_{ih}(V_{ih}) \cap X_{jh}(V_{jh}) \neq \emptyset$, then the map $X_{ih}^{-1} \circ X_{jh} : X_{jh}^{-1}(U_{ij}) \rightarrow X_{ih}^{-1}(U_{ij})$ is bijective, continuous and piecewise linear. The discrete surface is defined by

$$\mathcal{M}_h = \bigcup_{i \in I} X_{ih}(V_{ih}).$$

The advantage of this way to discretize the surface $\mathcal{M}$ is that it allows the approximation of immersions.

I.3 The Isoperimetric Problem and the Cheeger Constant [Ben16, Sec. 3]

Let $\mathcal{M}$ be an n -dimensional, Riemannian manifold with $\text{vol}_n(\mathcal{M}) < \infty$. For a positive n -Hausdorff measure subset $A \subseteq \mathcal{M}$ having $(n-1)$ -Hausdorff measurable boundary ∂A , the **isoperimetric ratio** of A is given by

$$h^*(A) \stackrel{\text{d\'ef.}}{=} \frac{\text{vol}_{n-1}(\partial A)}{\text{vol}_n(A)}. \quad (\text{I.3.1})$$

The isoperimetric ratio is well-defined when A is a smooth submanifold with boundary of $\mathcal{M}$ or, more generally, when A is an integral current. We also define the **isoperimetric constant** of a subset $A \subseteq \mathcal{M}$ to be

$$\bar{h}(A) = \inf_{D \subseteq A} \frac{\text{vol}_{n-1}(\partial D)}{\text{vol}_n(D)}, \quad (\text{I.3.2})$$

where D ranges over all subsets of A which have smooth boundary.

For $t \in]0; \text{vol}_n(\mathcal{M})[$, the isoperimetric problem of volume t studies the problem of finding the subset A of $\mathcal{M}$ with $\text{vol}_n(A) = t$ and

$$\text{vol}_{n-1}(\partial A) = \inf \{ \text{vol}_{n-1}(\partial B) \mid \text{vol}_n(B) = t \}.$$

We call such an A an **isoperimetric minimizer**. For an in-depth exposition on the isoperimetric problem for Riemannian manifolds, see [Ros05].

The Cheeger constant is related to the isoperimetric problem in that we wish to minimize the isoperimetric ratio among all isoperimetric problems corresponding to volumes $t \in]0; \text{vol}_n(\mathcal{M})/2]$. We will call a set D in $\mathcal{M}$ with $h^*(D)$ well-defined and $\text{vol}_n(D) \leq \text{vol}_n(\mathcal{M})/2$ a **Cheeger candidate**.

As described by [Ros05], one obtains the **fundamental existence and regularity theorem** for isoperimetric minimizers of Riemannian manifolds.

Théorème I.3.1 (Fundamental Existence and Regularity). *When $\mathcal{M}$ is a **smooth, compact**, possibly with boundary, and $t \in]0; \text{vol}_n(\mathcal{M})[$, there exists a compact region $A \subset \mathcal{M}$ such that ∂A minimizes the $(n-1)$ -volume among all regions of n -volume t . Further, for any such minimizer, ∂A is a smooth embedded hypersurface of constant mean curvature, except for a singular set of Hausdorff dimension at most $n-8$. Finally, if $\partial \mathcal{M} \cap \partial A \neq \emptyset$, then $\partial \mathcal{M}$ and ∂A meet orthogonally.*

As suggested by I.3.1 and smooth approximation techniques, it is equivalent to minimize the isoperimetric ratio over smooth submanifolds with boundary or the more general collection of rectifiable currents to find the Cheeger constant. In the same way that we have defined isoperimetric minimizers, we can define minimizers for the Cheeger constant, which we refer to as Cheeger minimizers for brevity. Specifically, a **Cheeger minimizer** is a subset A of $\mathcal{M}$ such that $\text{vol}_n(A) \leq \text{vol}_n(\mathcal{M})/2$ and $h^*(A) = h(\mathcal{M})$. When $\mathcal{M}$ is compact, Buser showed that there exist volume(s) $t \in]0; \text{vol}_n(\mathcal{M})/2]$ so that Cheeger minimizers coincide with solutions to the isoperimetric problem of n -volume t in $\mathcal{M}$ [Bus82, Remark 3.3, Lemma 3.4]. Therefore, the existence and regularity of a Cheeger minimizer follows from I.3.1. Specifically, Buser proved:

Lemme I.3.2 (Buser, [Bus82, Lemma 3.4]). *Let $\mathcal{M}$ be an n -dimensional, compact Riemannian manifold, possibly with boundary. Then there exists $t^* \in]0; \text{vol}_n(\mathcal{M})/2]$ and a sequence of submanifolds $A_k \subset \mathcal{M}$ with smooth boundary such that $\text{vol}_n(A_k) = t^*$ for all k and*

$$h(\mathcal{M}) = \lim_{k \rightarrow \infty} \frac{\text{vol}_{n-1}(\partial A_k)}{\text{vol}_n(A_k)}.$$

Combining I.3.1 and I.3.2, we can immediately conclude that **Cheeger minimizers exists for all compact manifolds.**

I.4 Graphes

[Chu, Sec. 1.2] : In a graph G , let d_v denote the degree of the vertex v . We first define the Laplacian for graphs without loops and multiple edges. To begin, we consider the matrix L , defined as follows:

$$L(u, v) = \begin{cases} d_v & \text{if } u = v \\ -1 & \text{if } u \text{ and } v \text{ are adjacent,} \\ 0 & \text{otherwise.} \end{cases}$$

Let T denote the diagonal matrix with the (v, v) -th entry having value d_v . The *Laplacian* of G is defined to be the matrix

$$\mathcal{L}(u, v) = \begin{cases} 1 & \text{if } u = v \text{ and } d_v \neq 0 \\ -\frac{1}{\sqrt{d_u d_v}} & \text{if } u \text{ and } v \text{ are adjacent,} \\ 0 & \text{otherwise.} \end{cases}$$

We can write

$$\mathcal{L} = T^{-1/2} L T^{-1/2}$$

with the convention $T^{-1}(v, v) = 0$ for $d_v = 0$. We say v is an isolated vertex if $d_v = 0$. A graph is said to be nontrivial if it contains at least one edge.

$\mathcal{L}$ can be viewed as an operator on the space of functions $g: V(G) \rightarrow \mathbb{R}$ which satisfies

$$\mathcal{L}g(u) = \frac{1}{\sqrt{d_u}} \sum_{v \sim u} \left(\frac{g(u)}{\sqrt{d_u}} - \frac{g(v)}{\sqrt{d_v}} \right).$$

When G is k -regular, it is easy to see that

$$\mathcal{L} = I - \frac{1}{k} A,$$

where A is the adjacency matrix of G and I is an identity matrix. All matrices here are $n \times n$ where n is the number of vertices of G .

For a general graph, we have

$$\begin{aligned} \mathcal{L} &= T^{-1/2} L T^{-1/2} \\ &= I - T^{-1/2} A T^{-1/2}. \end{aligned}$$

(lien avec la théorie de l'homologie)

Since $\mathcal{L}$ is symmetric, its eigenvalues are all real and non-negative. We can use the variational characterizations of those eigenvalues in terms of the Rayleigh quotient of $\mathcal{L}$. Let g denote an arbitrary function which assigns to each vertex v of G a real value $g(v)$. We can view g as a column vector. Then

$$\begin{aligned} \frac{\langle g, \mathcal{L}g \rangle}{\langle g, g \rangle} &= \frac{\langle g, T^{-1/2} L T^{-1/2} g \rangle}{\langle g, g \rangle} \\ &= \frac{\langle f, Lf \rangle}{\langle T^{1/2} f, T^{1/2} f \rangle} \\ &= \frac{\sum_{u \sim v} (f(u) - f(v))^2}{\sum_v f(v)^2 d_v} \end{aligned} \tag{I.4.1}$$

where $g = T^{1/2}f$ and $\sum_{u \sim v}$ denotes the sum over all unordered pairs $\{u, v\}$ for which u and v are adjacent. Here $\langle f, g \rangle = \sum_x f(x)g(x)$ denotes the standard inner product in $\mathbb{R}^n$. The sum $\sum_{u \sim v} (f(u) - f(v))^2$ is sometimes called the *Dirichlet sum* of G and the ratio on the left-hand side of I.4.1 is often called the *Rayleigh quotient*.

From equation I.4.1, we see that all eigenvalues are non-negative. In fact, we can easily deduce from equation I.4.1 that 0 is an eigenvalue of $\mathcal{L}$. We denote the eigenvalues of $\mathcal{L}$ by $0 = \lambda_0 \leq \lambda_1 \leq \dots \leq \lambda_{n-1}$. The set of the λ_i 's is called the *spectrum* of $\mathcal{L}$. Let $\mathbf{1}$ denote the constant function which assumes the value 1 on each vertex. Then $T^{1/2}\mathbf{1}$ is an eigenfunction of $\mathcal{L}$ with eigenvalue 0. Furthermore,

$$\lambda_G = \lambda_1 = \inf_{f \perp T\mathbf{1}} \frac{\sum_{u \sim v} (f(u) - f(v))^2}{\sum_v f(v)^2 d_v}. \quad (\text{I.4.2})$$

The corresponding eigenfunction is $g = T^{1/2}f$ as in I.4.1.

The above formulation for λ_G corresponds in a natural way to the eigenvalues of the Laplace–Beltrami operator for Riemannian manifolds:

$$\lambda_M = \inf_{\int_M f = 0} \frac{\int_M |\nabla f|^2}{\int_M |f|^2}.$$

We remark that the corresponding measure here for each edge is 1. The measure for each vertex is the degree of the vertex.

We note that I.4.2 also has the formulation

$$\lambda_1 = \inf_f \frac{\sum_{u \sim v} (f(u) - f(v))^2}{\sum_v (f(v) - \bar{f})^2 d_v},$$

where

$$\bar{f} = \frac{\sum_v f(v) d_v}{\text{vol } G},$$

and $\text{vol } G$ denotes the volume of the graph G , given by $\text{vol } G = \sum_v d_v$.

[Ji+10, Sec. 2] The normalized Laplacian $\mathcal{L}$ is defined by

$$\mathcal{L} = I - D^{-1/2} A D^{-1/2}.$$

The spectral gap λ_G is the least nonzero eigenvalue of $\mathcal{L}$. The value λ_G can be expressed as the infimum of the Rayleigh quotient:

$$\lambda_G = \inf_g R(g) = \inf_g \frac{\sum_{u \sim v} (g(u) - g(v))^2}{\sum_u g(u)^2 d_u}$$

where g ranges over functions defined on the vertices of G satisfying $\sum_u g(u) d_u = 0$ (see [Chu]).

For a set S , the volume of S , denoted by $\text{vol}(S)$ is defined to be $\text{vol}(S) \stackrel{\text{def.}}{=} \sum_{v \in S} d_v$. The volume of G is written as $\text{vol}(G) = \sum_v d_v$. For a set S , the Cheeger ratio of S is defined to be

$$h_S \stackrel{\text{def.}}{=} \frac{|\partial S|}{\min\{\text{vol}(S), \text{vol}(G) - \text{vol}(S)\}},$$

where the boundary of S is denoted $\partial S \stackrel{\text{def.}}{=} \{\{u, v\} \in E \mid u \in S \text{ and } v \notin S\}$. Let $\bar{S}$ denote the complement of S . Then $h_S = h_{\bar{S}}$. The Cheeger constant of G is denoted by

$$h_G \stackrel{\text{def.}}{=} \min_S h_S.$$

Théorème I.4.1 (Cheeger inequality [Ji+10, Theo. 1]). *In a graph G , the Cheeger constant h_G and the spectral gap λ_G are related as follows:*

$$2h_G \geq \lambda_G \geq \frac{\alpha_G^2}{2} \geq \frac{h_G^2}{2}$$

where α_G is the minimum Cheeger ratio of subsets S_i consisting of vertices *with the largest i values in the eigenvector associated with λ_G , over all i .*

For a proof of the Cheeger inequality using eigenvectors, see [Ji+10, Sec. 3].

[Ji+10, p. 7] Since $\sum_v g(v) d_v = 0$,

$$\sum_v g(v)^2 d_v = \min_c \sum_v (g(v) - c)^2 d_v.$$

I.5 Plan

Théorème I.5.1 (Objectif). Soit $\mathcal{T}$ un maillage de Ω . Il existe une constante C , ne dépendant que de p , du diamètre de Ω et de $\text{reg}(\mathcal{T})$, telle que pour tout $u^T \in \mathbb{R}^T$ et tout $g \in W^{1-\frac{1}{p},p}(\partial\Omega)$, on a

$$\|u^T\|_{L^p} \leq \|u^{\mathfrak{M}}\|_{L^p} + \|u^{\mathfrak{M}^*}\|_{L^p} \leq C \left(\|\nabla_{\mathbb{P}_m^T}^T u^T\|_{L^p} + \|g\|_{W^{1-\frac{1}{p},p}(\partial\Omega)} \right).$$

$$\|\Pi^{\mathcal{M}} u^T\|_{L^p(\mathcal{M})} \lesssim \|\nabla^T \Pi^{\mathcal{M}} u^T\|_{L^p(\mathcal{M})}$$

$$\sum_{\kappa \in \mathfrak{M}} m(\kappa) u_{\kappa}^2 \leq C \sum_{\mathcal{D} \in \mathfrak{D}} m(\mathcal{D}) |\nabla^{\mathcal{D}} u|^2$$

- La courbure de la surface est minorée et donc dès que h est suffisamment petit, tous les triangles sont dans un voisinage tubulaire de la surface qui est tel qu'on peut toujours projeter de manière unique dessus (de tout point du maillage, on peut projeter). On peut projeter les maillages sur la surface et on peut le faire de sorte que les volumes restent en h^2 des triangles et les longueurs des arêtes restent en h et peut être que les angles soient minorés par une constante uniforme indépendamment de h (on peut l'avoir avec la contrainte sur la régularité du maillage).
- Les diamants doivent aussi être proches de la surface (il faudra projeter les diamants et estimer)
- Le u est de deux sortes: P^0 du maillage dual.
- On veut une inégalité comme:

$$\|\tilde{\nabla} u\|_{L^2} \geq C \|u\|_{L^2}$$

- Tout le but de l'opération est de montrer que

$$\sum_{i \sim j} (u_i - u_j)^2 \geq C h^2 \sum (u_i - c)^2 \quad (\text{I.5.1})$$

Il faut minimiser sur c , prendre la meilleure constante, la projection de u sur les constantes.

- **I.5.1** est une inégalité de graphe, c'est le laplacien sur les graphe qui relie le membre de gauche au membre de droite avec une constante qui est la plus petite valeur propre du laplacien sur le graphe. **Il faut estimer cette constante.**
- Ceci est vrai avec $h = h(G)$ (Cheeger) avec

$$h(G) \stackrel{\text{déf.}}{=} \inf_{|S| \leq \frac{|T_h|}{2}} \frac{|\partial S|}{|S|}$$

- Il y a autant d'arêtes qui relie S aux autres que d'arêtes de bord qui contiennent les points de S .
- Tout les boulot est de montrer que

$$h(G) \geq h c$$

- Soit S . On considère l'ensemble de tous les triangles qui sont de centre un point de S et on projette ces triangles sur la surfaces.
- On considère le sous-ensemble de la surface qui est le projeté des triangles du maillage pour lesquels il y a un point de S qui est dedans. On appelle $A \subset \mathcal{M}$ cet ensemble.

$$A = \bigcup_{i \in S} \Delta_i^{\mathcal{M}}.$$

- D'après les lemmes de projection et les hypothèses : $|\partial S| \sim \frac{|\partial A|}{h}$ et $|S| \sim \frac{|A|}{h^2}$ et donc

$$\frac{|\partial S|}{|S|} \sim \frac{|\partial A|}{|A|} h$$

- $\frac{|\partial A|}{|A|}$: on est sur la surface, c'est la constante d'isopérimétrie de la variété et c'est minoré par quelque chose de strictement positif avec la condition $|A| \leq \frac{|\mathcal{M}|}{2}$.

I.5.1 État de l'art

I.6 Démonstration

Lemme I.6.1. Cf I.2.5.

$$\frac{m(\kappa)}{m(\Pi^{\mathcal{M}} \kappa)} = 1 + \mathcal{O}(h^2).$$

Proposition I.6.2 (Une inégalité de graphe). Soit $u: V(G) \rightarrow \mathbb{R}$ où G est le graphe associé à un maillage $\mathcal{T}$ tel que $\text{reg } \mathcal{T} \dots$. Alors

$$\sum_{i \sim j} (u_i - u_j)^2 \geq \min_c C h^2 \sum_i (u_i - c)^2. \quad (\text{I.6.1})$$

Définition I.6.3. Constante isopérimétrique

Lemme I.6.4. L'inégalité I.6.1 est vraie pour $h = h(G)$.

■ Démonstration.

□

Définition I.6.5. Soit $\mathcal{T}$ un maillage DDFV. On lui associe un graphe $G = (V, E)$ avec

$$V \stackrel{\text{déf.}}{=} \{x_\kappa \mid \kappa \in \mathfrak{M}\}.$$

On applique la proposition I.6.2 à une fonction scalaire primale $u = (u_\kappa)_{\kappa \in \mathfrak{M}}$.

$$U^\top U = \sum_{i \in V} u_i^2.$$

D'une part,

$$\|u\|_{L^2}^2 = \sum_{\kappa \in \mathfrak{M}} m(\kappa) u_\kappa^2 = \sum_{i \in V} \underbrace{m(\kappa)}_{\sim h^2} u_i^2 \sim h^2 U^\top U$$

D'autre part, pour $i \sim j$,

$$\begin{aligned} (u_i - u_j)^2 &= \left| \tilde{\nabla}_{i,j} u \cdot (x_j - x_i) \right|^2 \\ &\lesssim h^2 \left| \tilde{\nabla}_{i,j} u \right|^2 \\ &\lesssim \int_{D_{i,j}} \left| \tilde{\nabla}_{i,j} u \right|^2 \\ \sum_{i \sim j} (u_i - u_j)^2 &\lesssim \int_{D_{i,j}} \left| \tilde{\nabla}_{i,j} u \right|^2 = \|\nabla u\|_{L^2}^2. \end{aligned}$$

On veut $\|u\|_{L^2}^2 \lesssim \|\nabla u\|_{L^2}^2$. A-t-on $h^2 U^\top U \lesssim U^\top L U$? (U de moyenne nulle).

Lemme I.6.6. Soit $A \subset \mathcal{M}$,

$$\inf \frac{|\partial A|}{|A|} \dots$$

Lemme I.6.7.

$$\frac{|\partial S|}{|S|} \sim \frac{|\partial A|}{|A|} h$$

Proposition I.6.8 (Relaxation condition de Cheeger). Pour $0 < k < 1$, on définit

$$h_k(\mathcal{M}) \stackrel{\text{déf.}}{=} \inf_{\substack{A \subset \mathcal{M} \\ |A| \leq k|\mathcal{M}|}} \frac{|\partial A|}{|A|}.$$

Alors

$$\begin{cases} h_k(\mathcal{M}) = h(\mathcal{M}) & \text{si } \frac{1}{2} \leq k < 1 \text{ (relaxation),} \\ h_k(\mathcal{M}) \geq h(\mathcal{M}) & \text{si } 0 < k < \frac{1}{2} \text{ (restriction).} \end{cases}$$

■ **Démonstration.** On remarque que pour $0 < k \leq k' < 1$, alors $h_{k'}(\mathcal{M}) \leq h_k(\mathcal{M})$ car on minimise sur un ensemble plus grand.

$$h(\mathcal{M}) = h_{1/2}(\mathcal{M})$$

□

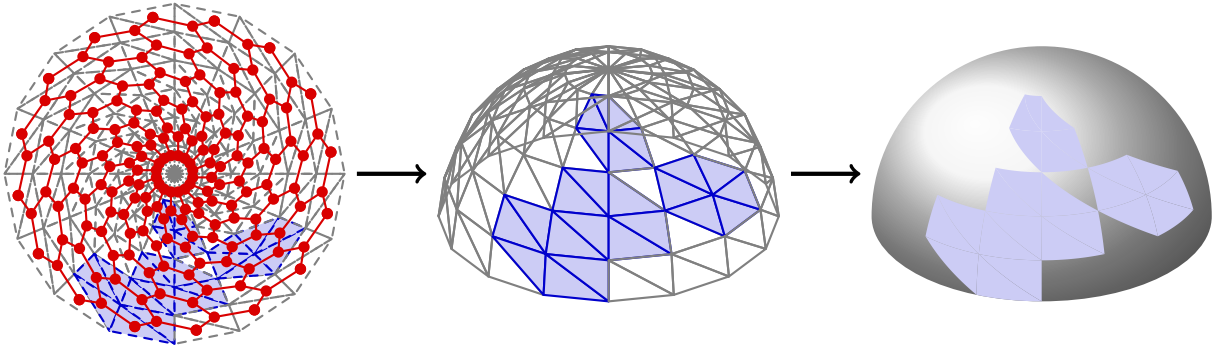


FIGURE I.14. maillage non régulier

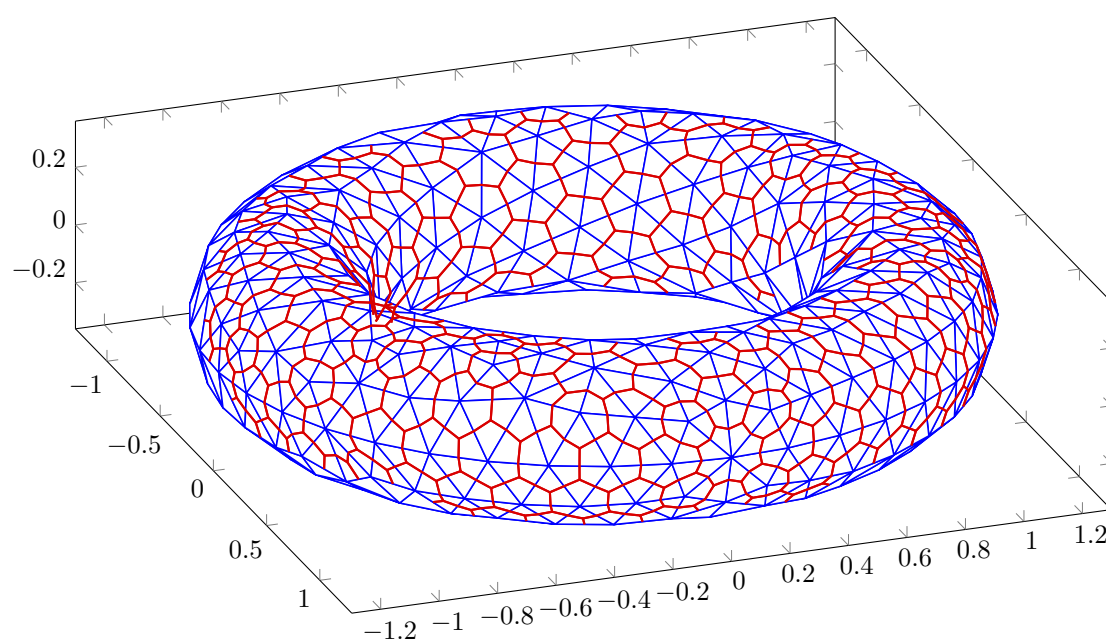


FIGURE I.7. Polyédrique

DEUXIÈME PARTIE

Appendices

Annexe A

Simplicial Complexes

<https://team.inria.fr/geometrica/files/2015/11/1-simplicial-complexes.pdf>

A.1 Geometric simplices

Définition A.1.1. A k -simplex σ is the convex hull of $k + 1$ points of $\mathbb{R}^d$ that are affinely independent.

$$\sigma = \text{conv}(p_0, \dots, p_k) = \left\{ x \in \mathbb{R}^d, x = \sum_{i=0}^k \lambda_i p_i, \lambda_i \in [0; 1], \sum_{i=0}^k \lambda_i = 1 \right\}$$

$k = \dim(\text{aff}(\sigma))$ is called the dimension of σ .

1-simple = line segment, 2-simplex = triangle, 3-simplex = tetrahedron

$V(\sigma)$ = set of vertices of a k -simplex σ

$\forall V' \subset V(\sigma)$, $\text{conv}(V')$ is a face of σ .

A.1.1 Geometric simplicial complexes

A finite collection of simplices K called the faces of K such that

- $\forall \sigma \in K$, σ is a simplex
- $\sigma \in K, \tau \subset \sigma \implies \tau \in K$
- $\forall \sigma, \tau \in K$, either $\sigma \cap \tau = \emptyset$ or $\sigma \cap \tau$ is a common face of both

The dimension of a simplicial complex K is the max dimension of its simplices.

The underlying space $|K| \subset \mathbb{R}^d$ of K is the union of the simplices of K .

A.1.2 Triangulation of a finite point set of $\mathbb{R}^d$

- A simplicial d -complex K is pure if every simplex in K is the face of a d -simplex.
- A triangulation of a finite point set $P \in \mathbb{R}^d$ is a pure geometric simplicial complex K s.t. $\text{vert}(K) = P$ and $|K| = \text{conv}(P)$

A.1.3 Triangulation of a polygonal domain of $\mathbb{R}^2$

A triangulation of a polygonal domain $\Omega \in \mathbb{R}^2$ is a pure geometric simplicial complex K s.t. $\text{vert}(K) = \text{vert}(\Omega)$ and $|K| = \Omega$

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